

THRESHOLD

Cross it, and the place speaks.

Archival heritage records contain thousands of historically significant Indian sites missing from modern maps and tourism platforms. Our system extracts, maps and contextualizes these forgotten sites, surfacing them for expert verification and for visitors on the ground.

SMART INDIA HACKATHON 2026 / INTERNAL ROUND

Theme: Heritage & Culture

Team Sigmoid / JSS University

This document is the single source of shared context. Everyone on the team reads it before writing anything.

SECTION 01

The problem

Between 1916 and 1922 the Archaeological Survey of India published a monument by monument survey of Delhi Province. It recorded roughly **1,300 structures**. Each entry described what the structure was, when it was built, and, in the prose of the period, where it stood.

Today the ASI Delhi Circle centrally protects about **174 monuments**.

The difference is not a story about demolition. A large share of that record simply stopped being *findable*. Structures stand unmarked in residential colonies, beside arterial roads and inside parks, with nothing to identify them, no presence on any tourism map, and no explanation available to anyone who walks past.

These two numbers are load-bearing and still need verifying

They open the pitch, they appear twice in the problem statement, and they are currently sourced from search results rather than from the primary documents. Task T6.1 sends someone to count Zafar Hasan's entries and find ASI's published list. If either number is wrong, that is the most valuable thing anyone finds this week.

Three connected failures

- 1. The record is digitised but not actionable.** The surveys have been scanned and are public. But they exist as page images and unstructured prose written in period units and period place names: "*about two hundred yards to the north-west of the Kotla*", "*half a kos beyond the old gate*". No modern mapping, tourism or conservation system reads them.
- 2. Nobody knows what is missing.** Because the historical record has never been reconciled against modern maps and protection lists, there is no working inventory of what is represented today and what is not. Attention flows to the same small set of famous sites, not because the others lack significance, but because no one has a list.
- 3. Heritage cannot explain itself on site.** Even at a well known monument, a visitor standing inside the complex cannot tell which structure they are looking at without a paid guide. Interpretation is signboards and human guides, both limited in language, depth and availability. The crafts, rituals and performance traditions attached to these places are almost entirely absent.

Our problem statement

RECONNECTING INDIA'S ARCHIVAL HERITAGE RECORDS WITH ON-GROUND VISITOR EXPERIENCE

Written from the stakeholder's point of view (ASI / Ministry of Culture, Software, Heritage & Culture). Full text in `plan/problem-statement.md`. It is deliberately solution-agnostic: three different teams could answer it three different ways. Say plainly that the team framed it. Never present it as an ASI-issued document.

SECTION 02

What we are building

Two engines over one heritage graph.

DISCOVERY

Read the archival record. Pull out every structure a page mentions. Convert its written location into a modern coordinate with an honest uncertainty radius. Check whether anything on today's map sits inside that radius. If nothing does, we have surfaced a **Representation Gap**: something the record knew about that no modern map shows.

EXPERIENCE

A visitor walks a heritage site. When they are close to a structure, facing it, and have stood still for three seconds, it begins telling them about itself. No search, no typing, no guide, no knowing in advance what they are looking at.

The angle, in one sentence

Audio guides tell you about places someone already decided were worth telling you about. Heritage digitisation produces archives nobody is standing in front of. THRESHOLD is the only one where **the archive decides what the tour contains**. The discovery engine is not a side feature. It is the content pipeline for the tour.

Success is one unbroken journey

```
a 1916 Page names a structure
|
v  its Spatial Clue projects to a coordinate + Uncertainty Radius
|
v  nothing in the Modern Baseline sits inside it
|
v  so it is a Representation Gap
|
v  a Reviewer verifies it
|
v  it appears in a Visitor's Hidden Heritage panel
|
v  the Visitor walks to it, faces it, waits three seconds
|
v  it speaks, and every sentence traces back to that Page
```

That sentence is the whole product. Everything marked MVP exists to make one instance of it real, end to end, on stage. Anything that does not serve it is V2, stretch, or cut.

The one principle

Show the evidence, or don't show it at all.

Every pin, fact, confidence number and status must expose where it came from within one interaction. This single rule kills fabricated visitor charts, generated "historical reconstructions", unsourced narration, and any pin we cannot defend. It turns the project's biggest risk, a model confidently inventing history, into its stated discipline.

If a proposed feature cannot show its evidence, it does not get built. No exceptions, and none "just for the demo".

SECTION 03

Shared vocabulary

These words mean exactly one thing each. Six people merging work is where loose synonyms turn into real confusion, so use these and not their cousins. Full glossary in `CONTEXT.md`.

Two phrases that must never appear, anywhere

"**discovered a monument**" - nothing here discovers anything. The archive recorded it, we projected it, a Reviewer confirms it. Say *surfaced a Candidate* or *identified a Representation Gap*.

"**AI found**" - the extraction reads a Page. Attribute claims to the Volume, not to the model.

This is not pedantry. A judge will attack any claim that an AI decided what heritage is, and the vocabulary is how we stay out of that argument entirely.

The place

Term	Means	Do not say
Heritage Site	A bounded place a visitor travels to, holding one or more Heritage Points.	monument, attraction, destination
Heritage Point	A single named structure inside a Site that can speak for itself. The smallest thing with its own story.	POI, marker, node, landmark
Zone	The traced footprint of a Heritage Point. The ground it actually occupies.	geofence, boundary, area
Approach Ring	The band around a Zone within which a Point is close enough to speak. Derived from the Zone, never drawn by hand.	radius, buffer

The visitor

Term	Means	Do not say
Visitor	The person moving through a Site. Anonymous. We know where they are, never who they are.	user, tourist, customer
Facing	Whether the Visitor's heading points at a Point's centre within tolerance. Standing beside it with your back turned is not Facing it.	orientation, bearing
Dwell	Unbroken time holding every trigger condition. Walking past is not Dwelling.	wait, delay, linger
Threshold Crossing	The moment a Visitor is inside an Approach Ring, Facing the Point, and has Dwelled long enough. The event the product is named for.	entry, trigger, arrival
Walk	One Visitor's continuous session through one Site.	tour, trip, session
Location Source	Where a position comes from. Simulated and real travel the same path; only the source differs.	GPS, demo mode, fake location

The telling

Term	Means	Do not say
Fact Sheet	The sourced, checkable record of what is true about a Point. Every Narration derives from one. Nothing is ever said that a Fact Sheet does not support.	content, description, info
Narration	One telling of one Point, for one Persona, in one language.	audio, clip, script
Persona	Who a Narration is written for. Changes how a Point is told, never which Points exist.	profile, mode
Interest Tag	A subject label used to decide whether a Point belongs in a Route. Independent of Persona: a child still wants the military points.	category, theme, topic
Living Tradition	The intangible practice a Point holds. A ritual, craft, cuisine or performance. Heritage that is done, not built.	culture, custom
Route	An ordered selection of Points fitting a Visitor's Interest Tags and time budget.	itinerary, trail, plan

The record

Term	Means	Do not say
Volume	A scanned archival publication we have ingested whole.	document, book, corpus
Page	One leaf of a Volume, held as both scanned image and text.	scan, sheet, folio
Mention	A passage naming a structure. A claim the archive made, not one we make.	entity, hit, detection
Spatial Clue	The written description of where a Mention's structure stands, relative to an Anchor.	location, geo hint
Anchor	A present-day landmark whose position we know, from which a Spatial Clue is measured.	reference point, origin
Uncertainty Radius	The distance within which a projected structure is believed to lie. Always shown. A coordinate without one is a lie.	error, margin, accuracy
Candidate	A structure projected from the record, unconfirmed against today's map. The furthest anything automated may go.	discovery, find, new site
Modern Baseline	Today's record of what is already mapped, against which a Candidate is checked.	current data, ground truth
Representation Gap	A Candidate with nothing from the Baseline inside its radius. Recorded once, unmapped now. The thing this product exists to surface.	lost monument, hidden site
Confidence	A decomposable score whose parts are always visible. Never a verdict, never a state.	probability, certainty
Evidence	The Page, passage, Anchor and Baseline check behind a Candidate. Reachable in one step, always.	proof, provenance
Reviewer	The only actor who can move a Candidate out of candidate.	admin, expert, moderator

SECTION 04

The five decisions

Recorded in docs/adr/. These are settled. Reopening one costs more than it saves, so if you disagree, raise it before building, not after.

ADR-0001

One Next.js app, no separate backend service

The blueprint called for a Python service because the archival pipeline is Python-shaped. Rejected: seven days, one developer, and a second runtime buys two deploys, CORS, cold starts and a second thing to debug under pressure. Everything at request time is TypeScript in Next.js route handlers on Vercel. The genuinely Python-shaped work (OCR, page download, audio rendering) is not request-time work. It runs once on a laptop and its output is committed.

ADR-0002

Static content in the repo, database only for what changes

Sites, Points, Zones, Fact Sheets, Narrations and audio never change at runtime, so they are committed and read at build time. Postgres holds only Candidates, review state and Walk logs. Putting everything in Postgres reads better on an architecture slide but makes every screen depend on a free-tier database that sleeps, in exchange for no capability we need. Consequence worth naming: content edits are git commits and need a redeploy. Correct while we author the content, wrong the day someone else does.

ADR-0003

A Candidate is the ceiling of what automation may assert

Nothing automated may claim a monument exists. The pipeline reaches candidate and stops, however high the Confidence. Only a Reviewer goes further. We could have let a high score auto-promote, which would make the demo self-running. Rejected because the entire credibility of the project rests on not overclaiming, and because `matched_existing`, the state where the pipeline finds something already documented, is the most persuasive evidence that we are not inventing pins.

ADR-0004

Narration is written and rendered ahead of time

Every Narration is generated offline from a Fact Sheet, reviewed by a human, and rendered to an audio file that ships with the app. At a Threshold Crossing the app plays a file. Live generation is closer to the original vision but means seconds of silence at the exact moment a Visitor arrives, speech quality that depends on the device, and an unreviewed sentence about a real monument spoken aloud to a historian. The only live model surface is one optional question box grounded in a single Fact Sheet, where a wrong answer is visible and recoverable.

ADR-0005

A Threshold Crossing needs Facing and Dwell, not proximity

Proximity-only triggering is far simpler and is what the reference prototype did with 500m circles. It fails exactly where it matters: Heritage Points at Red Fort sit tens of metres apart, so a Visitor between Diwan-i-Aam and Rang Mahal satisfies both, and a Visitor walking past toward somewhere else gets interrupted. Facing disambiguates which structure a Visitor means; Dwell separates arriving from passing. The cost is a harsher failure mode, so the conditions must be visible on screen as they accumulate.

SECTION 05

Architecture

One repo. One deploy. One language. Anything that looks like a second service is a local script that runs on a laptop and commits its output.

```

run once on a laptop      scripts/ (tsx, never deployed)
output is committed      ingest-volume . render-audio
                           pull-baseline . seed-db . check-content
                           |
                           | writes files
                           v
+-----+
| content/ + public/      (in git, no network at runtime) |
| sites . points . zones . factsheets . anchors . baseline.geojson |
| public/audio/*.mp3 . public/pages/*.jpg . public/images/* |
+-----+
| imported at build time |
v
+-----+
| Next.js 16 app (Vercel) |
|                           |
| Client                   Server (route handlers) |
| - LocationProvider (sim|gps) - POST /api/discover/analyse --> Gemini
| - Location engine -> crossing - POST /api/ask -----> Gemini
| - Route planner          - POST /api/walk/crossing |
| - Narration player       - GET /api/candidates |
| - Leaflet map (CARTO tiles) - PATCH /api/candidates/:id |
| - Evidence panel |
+-----+
|
v
Supabase Postgres - mutable state only
candidates . candidate_events . walk_crossings

```

Every external dependency has a committed fallback

And the fallback is a first-class path exercised in development, not a catch block written on day seven. Gemini down means the cached result for that Page with a visible *cached* badge. Supabase paused means the committed seed file with a *stale* badge. A labelled fallback is a stronger position than a disguised one: if a judge asks whether it is live, the answer is already on the screen.

Locked stack, versions verified

Layer	Choice	Why this one
Framework	Next.js 16.3.1, React 19	Route handlers hide the API key. One deploy target.
Styling	Tailwind 4.3.3 tokens + Radix primitives	No component library. Their defaults look like every other hackathon project.
Map	Leaflet 1.9.4 + react-leaflet 5.0.0, CARTO tiles	No API key, no account, no billing. One CSS line makes it aged paper.
Geometry	@turf/turf 7.4.0	All spatial work in memory. No PostGIS.
Database	Supabase 2.112.3	Mutable state only. Three tables.
Model	@google/genai 2.17.1, gemini-3.7-flash	Free tier, about 10 requests per minute. Serial only.
Audio	Edge TTS, rendered offline	No key, no quota, real Indian neural voices.
Tests	Vitest 4.1.11, zod 4.4.3	Five pure modules only.

The seams that keep this navigable

- **All Supabase access lives in `lib/store/`.** Nothing else imports the client, so every table and column name exists in exactly three files.
- **All model access lives in `lib/ai/gemini.ts`.** The only reader of the API key, and server-only. If the key appears in a client bundle, someone broke this rule.
- **Pure logic does no I/O.** Engine, planner, resolver, baseline, confidence and transitions take data and return data. That is why they test in milliseconds and why the UI can explain a rule instead of surfacing an error.
- **Coordinates are `[lng, lat]` everywhere.** GeoJSON order, because turf demands it. Leaflet wants `[lat, lng]`. The flip happens in one helper and nowhere else. This single convention prevents the most common bug in this kind of app.
- **Content is data, not code.** A validation script checks every content file against a schema and runs in CI, so a malformed Fact Sheet fails the build rather than a Threshold Crossing on stage.
- **Nothing renders a claim without a route to its Evidence.** Enforced by the component's type, not by discipline.

SECTION 06

The core mechanic

This is the piece the whole demo turns on, and the one most likely to break quietly. Read it even if you are not building it.

The trigger rule

Narration begins when ALL THREE hold, continuously:

1. **INSIDE** the Visitor is within the Approach Ring
2. **FACING** bearing to the structure's centre is within +/- 60 degrees
of the Visitor's heading
3. **DWELL** both have held for 3 seconds without a break

Re-arming: leave the ring + 10m for 3 seconds

Firing: at most once per Walk, unless re-armed

Inside: crossing from the ring into the Zone polygon itself fires
a short detail Narration, once per Walk

Why each condition exists

- **Facing** disambiguates. Red Fort structures sit tens of metres apart. A Visitor standing between Diwan-i-Aam and Rang Mahal is inside both rings. Heading decides which one they actually mean.
- **Dwell** separates arriving from passing. Without it, walking toward your destination sets off every structure you pass.
- **Hysteresis on exit** stops GPS jitter re-firing the same Narration three times while a Visitor stands still.

One polygon, two behaviours

A Zone is hand-traced once from the structure's OSM footprint. The Approach Ring is generated from it by buffering 25 metres. It is never drawn by hand. That gives two behaviours from one authored piece of geometry: the buffer is "near it", the polygon is "inside it".

The failure mode to design against

A trigger that fires wrongly is survivable. A trigger that **never fires** is not, and this rule fails silently. So the conditions are drawn on screen as they accumulate: a facing cone from the Visitor marker, a three second ring filling, and the named reason when a condition fails ("facing away", "1.2s more"). That turns an invisible rule into the most legible thing on the screen, and it is also the fastest way to debug it. The tolerances are environment variables so they can be loosened from the Vercel dashboard at 2am without a rebuild.

Simulated and real are the same code path

One LocationProvider interface with two sources. Simulated mode lets you drag the marker, nudge heading with arrow keys, and auto-walk a Route. Real mode uses the browser geolocation and compass. That identity is the honest answer to "is this fake?", and it is why we can demonstrate a walk through Red Fort from a college auditorium and then flip to real GPS on a phone for fifteen seconds to prove the point.

If the compass is unavailable or denied, heading falls back to GPS course over ground. If the Visitor is stationary and there is no compass at all, the Facing gate **disables itself and says so** rather than blocking every Crossing.

SECTION 07

Features

MVP the demo fails without it. **V2** builds if MVP lands early. **STRETCH** never blocking. The MVP set sums to exactly the one journey in Section 02 and nothing else is MVP.

Visitor

#	Feature	Tag	Proves it works when
F1	Delhi heritage map	MVP	11 Sites render as pins on a sepia basemap; a card opens with name, period and Point count
F2	Tour planner	MVP	5 Interest Tags, 4 time budgets, 3 Personas; selecting nothing is valid and means everything
F3	Route generation	MVP	45 to 90 minutes visibly changes which Points are included and their order; deterministic
F4	Simulated location	MVP	Drag moves position; arrow keys rotate heading and step; auto-walk follows the Route
F5	Real GPS	MVP	The marker tracks real movement on a phone; heading degrades gracefully when no compass exists
F6	Threshold Crossing	MVP	Facing it for 3s fires. Walking past does not. Between two Points, only the faced one fires
F7	Narration + transcript	MVP	Audio plays, transcript highlights in sync; with the device muted the transcript alone carries it
F8	Trigger visualisation	MVP	A cone and a filling ring show why it fired, and name the reason when it did not
F9	Inside detail	MVP	Entering the Zone itself fires a short extra Narration, once per Walk
F10	Living Tradition card	MVP	Every Point shows a practice, 2 to 3 sentences, and whether it is living, dormant or lost
F11	Persona switching	MVP	The same Point replayed under a different Persona is audibly a different telling
F12	Ask about this place	MVP	Answers in under 6s citing Fact Sheet lines; declines rather than guessing when uncovered
F13	Hidden Heritage panel	MVP	Nearby under-represented places ranked, each with its evidence line; one from the pipeline
F14	Then vs Now	MVP	A draggable divider on 3 to 5 Points, each image credited with its licence
F15	Hindi Narration	V2	A toggle switches audio and transcript for all 11 Red Fort Points
F16	Deep sites 2 and 3	V2	Qutub and Humayun's Tomb each walk end to end. Zero code, pure content
F17	Offline Walk	STRETCH	With devtools offline after first load, Route, tiles and audio all still work

Researcher

#	Feature	Tag	Proves it works when
F18	Volume shelf and Page viewer	MVP	A real scanned 1916 page renders beside its text, served locally not from archive.org
F19	Live Mention extraction	MVP	Mentions stream in under 20s with passages highlighting; failure serves the cached result, labelled
F20	Spatial resolution	MVP	An uncertainty circle draws and contracts, then pins. An unknown Anchor shows as unresolvable, never guessed
F21	Modern Baseline check	MVP	Every Candidate reports matched_existing with a name, or a Representation Gap
F22	Evidence panel	MVP	One interaction from anywhere shows page, passage, anchor, maths, baseline and the confidence breakdown
F23	Multi-Volume cross-reference	V2	A Candidate supported by two Volumes shows both and its cross-source confidence rises

Reviewer

#	Feature	Tag	Proves it works when
F24	Candidate queue	MVP	Sorted, filterable, every row opens the same Evidence panel
F25	Verify / reject / match	MVP	A status change persists and the pin colour changes on the map. Illegal transitions are not rendered
F26	Walk log	MVP	The Walk performed on stage three minutes earlier appears in it
F27	Coverage summary	V2	Three real counters derived from actual rows. No charts of invented data

The Candidate lifecycle, exactly

extracted -> geo_resolved -> candidate -> under_review -> verified | rejected | matched_existing

Automation stops at candidate. Only a Reviewer moves it further. verified and rejected are terminal; matched_existing can reopen to under_review if a Reviewer disputes the match. Confidence is a number displayed beside a Candidate. It is never a state and never a promotion trigger.

SECTION 08

Who is building what

One developer on the core, five teammates on independent features. Each brief is standalone: build it as your own Next.js app, then submit it as a pull request into `contrib/`. Full briefs in `plan/teammate-briefs/`.

Brief	Feature	Time	Character of the work
core	Map navigation, location tracking, geofencing, the Threshold Crossing	-	The one piece that cannot slip
1	Discovery engine: archival page to pins on a map	2 to 3 days	Highest difficulty. Extraction, geometry, uncertainty
2	Authority dashboard: the Candidate review queue	1.5 to 2 days	Most database. Schema, state machine, audit trail
3	Route planner: interests and time to an ordered walk	1 to 1.5 days	Most algorithm. Scoring, budget, exact ordering
4	Narration: audio, transcript, personas, grounded Q&A	2 days	Most content. Fact sheets, TTS, grounding
5	Explore, Hidden Heritage, landing page	1.5 to 2 days	Most design. The visual identity lives here

A PR is accepted when three things hold

Every changed file sits inside your own `contrib/` folder. No `.env`, no API key, no credential in the diff. No `node_modules/`, no `.next/`, no media over 5MB.

That is the whole bar. Code style, architecture and whether the feature ships are not grounds for rejection. `contrib/` is excluded from the build, typecheck, tests and deploy, so nothing in it can break anything.

Say honestly what does not work yet. A PR that names what is broken is more useful than one that pretends.

How to submit

```
gh repo fork <MAIN_REPO_URL> --clone
cd threshold
git checkout -b brief-N-<name>-<yourname>
mkdir -p contrib/brief-N-<name>-<yourname>
# copy your app in, then
git diff --staged | grep -iE "api[_-]?key|secret|token" # must print nothing
git add contrib/ && git commit -m "contrib: ..." && git push -u origin HEAD
gh pr create --fill
```

Three rules that make five parallel builds mergeable

- **Copy** `lib/types.ts` **verbatim**. Do not rename a field, do not change a type, do not add one because it seemed useful.
- **Coordinates are [lng, lat]**. Flip to Leaflet order in one helper, nowhere else. This is the bug you are most likely to ship.
- **Run your brief's Verify section before opening the PR** and paste its output into the description. If something does not pass, say so in the PR rather than hiding it.

SECTION 09

Seven days

Every phase ends at something we could put in front of a judge if the clock stopped there. We never build a layer, we always build a slice.

Day	Ships	Definition of done	Risk it retires
0 eve	Foundations	The Vercel URL serves an aged-paper page with correct type, and tests pass	Deploy surprises on day 6
1	A place speaks	Open Red Fort, tap Diwan-i-Aam, hear real Indian-voiced narration with the transcript following	Audio pipeline, ingestion, map
2	The Threshold Crossing	Drag toward Diwan-i-Aam, turn to face it, wait 3s. It speaks on its own. Turn away first and it does not	The hardest logic, retired early
3	A whole Walk	Pick interests and 45 minutes, walk the Route end to end, switch to Kids mid-walk and hear a different telling	Content volume
4	The record speaks	Pick a Page never demoed before. A 1916 passage becomes a pin with an honest radius and a baseline verdict	The riskiest external dependency
5	The loop closes	Verify a Candidate, find it green in Hidden Heritage, walk to it, hear it speak. The whole journey unbroken	Integration between the engines
6	Make it defensible	Every failure mode triggered on purpose and survived. Two rehearsals, one with wifi off	Demo-day surprise
7	Buffer, then depth	Buffer first. Then ONE of: deep sites 2 and 3, or Hindi, or offline. Not two	Overreach

Why the order is what it is

The two genuinely hard things land on days 2 and 4, early enough that discovering they are harder than expected still leaves room to simplify them rather than cut them. Red Fort stays alone until day 3 because breadth before depth would give three mediocre sites instead of one convincing one.

Where separate work has to meet

- **Content and code, continuously.** A validation script checks every content file against a schema in CI. A malformed Fact Sheet fails the build, not a Threshold Crossing on stage.
- **Zone polygons and the engine, day 2.** A stub square works until the real trace lands. Do not block the engine on tracing.
- **Discovery and the tourist app, day 5.** A verified Candidate must render as a Site pin. This is the only place the two engines touch. It is one function, and it is scheduled on day 5 rather than day 7 for exactly that reason.
- **Audio and transcript, day 1.** Cues come from the render script. If narration text changes and the audio is not re-rendered, the transcript silently desynchronises.

SECTION 10

The demo

Roughly 8 minutes, rehearsed twice, once with wifi switched off. The slot length is not yet confirmed, so a 4 minute cut is rehearsed too.

Time	Screen	What happens	What the judge takes away
0:00	landing	"The ASI catalogued about 1,300 monuments in Delhi between 1916 and 1922. About 174 are centrally protected today. The rest did not all disappear. They stopped being findable."	The problem, in two real numbers
0:45	explore	Delhi map, pick Red Fort	Eleven sites on aged paper
1:00	plan	History and Architecture, 45 minutes, History persona. Press Begin tour	Personalisation that will visibly matter
1:30	tour	The Route draws	Seven of eleven Points, ordered
1:50	tour	Drag toward Diwan-i-Aam. Turn away first, nothing. Turn to face it, the cone lights, the ring fills, it speaks	The moment. Transcript follows along
3:00	tour	Step inside the Zone	A shorter detail Narration
3:20	tour	Living Tradition card, then the Then vs Now slider	The theme answered, and 1860 against today
4:00	tour	Switch to Kids, replay the same Point	Same place, genuinely different telling
4:30	tour	Ask: "why is this hall so wide?"	A live grounded answer citing its Fact Sheet
5:00	discover	"Where does any of this come from?" Let a judge pick the page	A real 1916 scan; phrases light up; cards land
5:45	discover	The radius contracts, the pin drops, the baseline check runs	"Nothing in today's map sits inside this circle"
6:30	authority	Move it to under review, then verified	A human in the loop, by design
7:00	explore	It is now in Hidden Heritage, green	The loop closes
7:20	phone	Open the URL on Android, flip to real GPS	The blue dot is you. Not a simulation
7:40	-	"The record already knew these places existed. THRESHOLD puts them back on the map, and then lets them speak for themselves."	Close

If something breaks on stage

Say what broke, use the labelled fallback, and keep going. The *cached* and *stale* chips exist precisely so that this becomes a demonstration of the design rather than an apology.

SECTION 11

Risks and what we do about them

Every high-risk bet has a fallback that still yields a demo. Full register in `plan/06-risks.md`.

Risk	Signal	Fallback
The Threshold Crossing never fires on stage medium / fatal	A Point stops firing after a Route change. Any engine test going red	The trigger visualisation names the failing condition on screen, so you diagnose live in seconds. Tolerances loosen from the Vercel dashboard with no rebuild. Every Point stays tappable, so the demo degrades from "it knows what I am looking at" to "tap to hear"
Content authoring becomes the bottleneck high / high	End of day 2 with fewer than 5 Fact Sheets done	Cut Red Fort from 11 Points to 6. Cut Personas before Points: 11 Points at 1 Persona beats 6 Points at 3, because Route quality is more visible than Persona switching
Gemini rate-limited or unreliable medium / high	429s during rehearsal; extraction exceeding 8s repeatedly	A cached result exists for every ingested Page, generated on day 4 before the screen was built. Served with a visible <i>cached</i> chip. Rehearse on pages you will not use on stage so demo-day quota is untouched
Grounded Q&A says something wrong medium / high	Any rehearsal answer containing a fact not in the Fact Sheet	Only that Point's lines are sent, each with an id, and citations are required. Any answer citing a non-existent id is discarded. If it misbehaves in rehearsal, remove the Ask box entirely; nothing depends on it
Venue has no usable internet medium / high	Obvious on arrival	A local production build, rehearsed once on day 6. Content, audio and geometry are all local. An untested fallback is not a fallback, so it gets one real rehearsal
Audio does not play medium / high	First Crossing silent in a fresh browser profile	The Begin tour button unlocks the audio context. For the room, the synchronised transcript carries the whole Narration visually. This is the second reason it exists
The two engines never connect medium / high	Day 5 and no verified Candidate has rendered on the map	The bridge is one function and it is scheduled on day 5, not day 7. If it slips, Hidden Heritage reads Candidates straight from the seed file. Same visible result, less machinery
Spatial resolution produces wrong pins medium / medium	A Candidate lands in the Yamuna or 4km from anything plausible	This is why the Uncertainty Radius is mandatory. A wide circle is an honest answer; a bare pin is not. A Mention with no Anchor shows as unresolvable with its passage, never at a guess
Scope creep back to the blueprint high / high	Any sentence beginning "it would be cool if"	The cut list and the one principle. Those ideas belong on the roadmap slide, where they cost nothing and still land. Pitching a future is free; building one is not

SECTION 12

What we are not building

Each of these is a real feature and a real trap. Saying no to them in advance is what makes the yes list achievable.

Cut, and staying cut

Not building	Because
Document upload of any kind	Unbounded input. A judge's PDF will break it live. The Page shelf is interactive enough
A separate Traditions module	The Living Tradition field answers the theme. A second content type dilutes the demo
Analytics charts	A chart of invented numbers is the least convincing object on any hackathon screen. Three real counters only
Auth, roles, login	No judge will ask you to log in. A known, documented, accepted gap
PostGIS	All spatial work is in-memory turf. An extension to accelerate queries we never run is pure cost
A second AI provider in the demo path	One tested fallback beats two half-tested ones
Live TTS or live narration generation	Silence at a Threshold Crossing is the worst failure this project has
Historical-map georeferencing	A multi-week problem wearing a one-day costume
Cities beyond Delhi	The Anchor table is Delhi-specific, and that is the point
Generated imagery of real monuments	Violates the one principle. Not a time question
React Native, a Python service, HF Spaces	One dev, seven days. One language, one repo, one deploy
Any abstraction with one caller	No provider registries, no plugin layers, no config system beyond env vars

Deferred, and the model is already shaped for it

Hindi (Narration is already keyed by language, so it is content only) . Deep sites 2 and 3 (depth is data, not schema) . Offline Walk . Multi-Volume cross-referencing (the Confidence formula already reserves a weight for it).

Pitched on the roadmap slide, never built

AR . Bluetooth and indoor positioning . National-scale heritage graph . Crowdsourced verification . Hotel, travel and guide integration . Offline nationwide packs. These are genuinely good ideas and they belong in the roadmap, where they cost nothing and still land.

If you catch yourself scope-creeping

The source blueprint for this project contains AR, offline nationwide packs, hotel integration, crowdsourced evidence and a national heritage graph. All of it is attractive. None of it is buildable this week. The test is the one principle: **can this feature show its evidence?** If not, it does not get built. If yes, it still has to beat something already on the MVP list.

SECTION 13

Questions a judge will ask

Prepared answers. If you are in the room, know these. Full set being drafted as task T6.4.

“How do you know the AI isn't making these up?”

Five things, all visible on screen without us saying a word. Every claim traces to a page image and an exact passage they can read. Every coordinate carries an uncertainty radius, drawn on the map. Confidence is five named components with weights, not one opaque number. Nothing automated goes past candidate; a human moves it further or it stays there forever. And some Candidates come back `matched_existing`, which proves the pipeline finds real things rather than producing pins.

“So the AI decides what counts as a heritage site?”

No. Automation stops at Candidate. The Reviewer screen is where a person decides, and the audit trail records who moved it and when. That is not a limitation we are apologising for. It is the design, and it is written into our problem statement as a requirement.

“How is this different from Google Maps?”

Google Maps shows you what is already mapped. Every entry in our Hidden Heritage panel carries a line saying where the record of it came from and that today's map does not have it. One of them was surfaced from a 1916 document by our own pipeline an hour ago. That is a different kind of object from a search result.

“Isn't this just an audio guide?”

An audio guide's content pipeline is a human writing a script for a monument already on the map. Ours is an archival document deciding what the tour contains. The discovery engine is not a side feature; it is where the content comes from.

“What does your confidence number actually mean?”

It is a weighted sum of five components, each shown with its value and weight: source reliability, clue specificity, anchor precision, cross-source agreement and modern evidence. Open the Evidence panel and the arithmetic is right there. A single opaque percentage would be worse than showing nothing.

“Did a historian check any of this?”

Every narration sentence traces to a named source visible in the app, and the model was never allowed to add a fact, only to rephrase supplied ones. Generation happens offline and is human reviewed, so a wrong sentence is an error we can find and fix, not a live hallucination we cannot predict.

“What happens when there's no signal at a heritage site?”

Content, audio and geometry are all bundled with the app rather than fetched, so the Walk itself does not need the network. The only networked surfaces are the extraction screen and the question box, and both have committed fallbacks that label themselves on screen.

“Is the walking demo real, or a simulation?”

Both, and they are the same code path with a different location source. Here it is on a phone with real GPS. We simulate on stage because we are not at Red Fort, and the simulation mode is labelled rather than disguised.

“Where did your images come from, and are you allowed to use them?”

Archival photography is public domain by age; modern images are Creative Commons with credit shown at point of use. There is an attributions page listing every image, volume and dataset with its licence. No generated imagery of real monuments exists anywhere in the project.

“Anyone can open your authority dashboard and verify a candidate.”

Correct, and it is documented as an accepted gap rather than hidden. Row-level security permits only status updates and event inserts, never deletes, and the event table is append-only so tampering is visible. Authentication is one Supabase email link and was deliberately not this week's hour.

SECTION 14

House rules

These apply to everyone, in code, comments, commit messages, documents and anything a user reads. Full version in `CLAUDE.md`.

Writing

- **No em dashes. No en dashes.** Use a simple hyphen with spaces around it, or restructure the sentence. This applies to UI copy and spoken narration too, where an em dash also makes text-to-speech pause strangely.
- Straight quotes only. No emoji in code, comments or commit messages.
- Plain words over grand ones. "Find" not "leverage". "Use" not "utilise".
- Comments explain **why**, never what. A comment that restates the code gets deleted.

Code

- **Simple and boring beats clever.** If two approaches work, take the one a tired person reads faster at 2am.
- **No abstraction until something is needed twice.** A single caller does not justify an interface.
- **No any, no @ts-ignore.** If the type is hard, the design is wrong.
- **Every external call has its fallback in the same function.** Not in a wrapper. It must be impossible to read the call without seeing what happens when it fails.
- Delete dead code immediately. Do not comment it out.

Testing

Test-driven on five pure modules only: the location engine, the route planner, the spatial resolver, the confidence score and the status transitions. UI, route handlers, database calls and scripts get no tests on purpose. Those five have interacting rules that will break silently on day 5 while someone is polishing something else. Everything else is glue, and testing glue on a seven day clock costs more than it returns.

Working agreement

- **Never invent a requirement.** If you are guessing, say so out loud.
- Say when someone is scope-creeping, including when it is the team lead.
- Keep a running list of cut features so they can come back if time allows.
- Build MVP first. Something demoable must exist at every point.
- Report honestly. If tests fail, show the output. If a step was skipped, say so.

Where everything lives

CONTEXT.md	the glossary and the scope cut line
CLAUDE.md	house rules, seams, PR policy
docs/adr/0001-0005	the five settled decisions
plan/00-overview.md	problem, principle, north star, the one journey
plan/01-features.md	27 features with acceptance criteria
plan/02-architecture.md	components, diagrams, API payloads, env vars
plan/03-data-model.md	ER, enums, tables, permissions, seed data
plan/schema.sql	runnable
plan/04-design-system.md	tokens, shell, component states, screens
plan/05-build-plan.md	seven phases and the minute-by-minute demo
plan/06-risks.md	13 risks, each with a fallback that still demos
plan/problem-statement.md	the PS in SIH format
plan/teammate-briefs/	five independent build briefs, one per person

THRESHOLD / Cross it, and the place speaks.